Two-Axis Computerised Antenna Radiation Pattern Measurement System

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Abstract. Hands-on experimentation is essential in RF and telecommunications education, yet access to instrumentation for characterising antenna performance, such as radiation patterns, is often limited due to the high cost and proprietary nature of commercial systems. While simulation software can estimate patterns during design, real-world environmental factors significantly affect performance. In this reference frame, this work presents the design and implementation of a custom-made mechatronic system able to measure antenna radiation patterns. By allowing real-time visualisation, this test rig promotes a deeper understanding of the antenna's behaviour in a classroom context. The device, which besides educational can also be used for research purposes, features two degrees of freedom, supports various antenna types, and has been validated using a custom-built 868 MHz Yagi-Uda antenna. Experimental results closely match the ones obtained from simulation software, and all design files and code will be publicly released to support replication and community-driven development.

Keywords: Antenna Radiation Pattern, Measurement System, Educational Methodologies, Mechatronics

1 Introduction

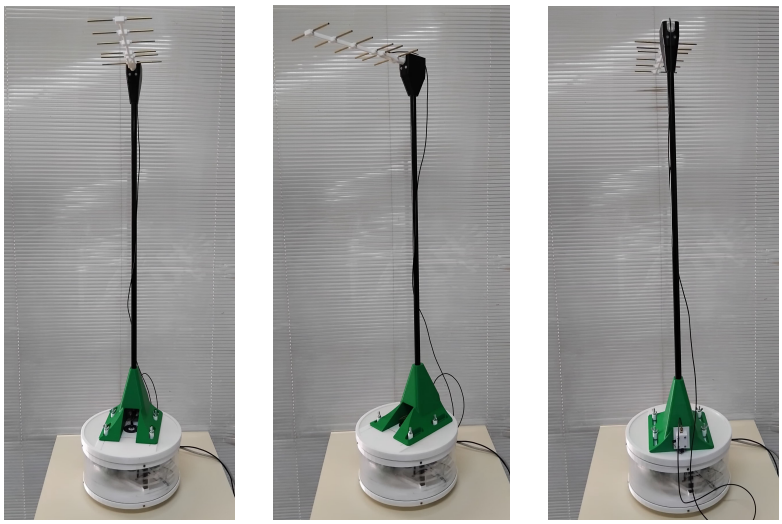
Characterisation of antenna radiation patterns is a fundamental aspect of wireless communication engineering, since the spatial distribution of radiated energy directly affects coverage, RF interference, and overall system performance [1]. While electromagnetic simulation tools are commonly employed to provide theoretical estimates of these patterns during antenna design, real-world measurements are indispensable to capture the effects caused by environmental factors and manufacturing tolerances, which simulations cannot fully replicate.

For those reasons, having access to measurement systems for antenna radiation patterns is a valuable tool for both research, product development and education applications. Regarding the latter, the use of such measurement setups in educational environments remains limited due to factors such as high acquisition costs, physical complexity, and frequently the need for specialised

infrastructure such as anechoic chambers. These limitations often restrict the ability of students to engage in hands-on experimentation, confining learning primarily to simulations and theoretical concepts. This gap can make the understanding of electromagnetic phenomena harder and reduce opportunities for experiential problem-solving.

To address these challenges, there has been growing interest in developing practical teaching tools. Various laboratory solutions and open-source projects have been proposed, aiming to reduce costs and complexity. Nevertheless, many of these approaches concentrate on specific frequency bands or lack the flexibility needed for broad didactic applications. Consequently, there remains a demand for accessible, modular, and low-cost platforms that can support experimental activities while illustrating the main antenna behaviour concepts.

In this context, the present work introduces the design and implementation of a computerised, two-axis radiation pattern measurement system, primarily intended to be used in teaching activities at the Polytechnic Institute of Bragança. The mechanical aspect of this measuring device is shown in Figure 1.



(a) Front view (b) Side view (c) Back view.

Fig. 1: The antenna radiation pattern measurement system.

The details behind the construction of this equipment will be presented later in Section 3. But first, an overview of existing similar measurement systems will be provided in Section 2, where both advantages and limitations of each solution will be highlighted. The result of this review and preliminary analysis allows this work to be positioned within the panorama of existing solutions. Section 4 will provide the experimental results obtained using a proof antenna. The close proximity between the experimental results and those generated by electromagnetic

simulation software will become evident, which validates the current approach. Finally, Section Finally, Section 5 will present the final conclusions of this work as well as some guidelines that will be followed in its future development.

2 State of the Art

The fundamental importance of antenna radiation pattern characterisation extends not only to wireless communication system optimisation but also to engineering education, where understanding the way those patterns emerge is critical to understanding signal propagation and antenna performance. However, the abstract nature of these electromagnetic concepts often makes them difficult to grasp, especially in the absence of practical visualisation tools. Those facts highlight the need for accessible experimental platforms that aid student learning by linking theory and practice.

Commercial antenna radiation pattern measurement systems, exemplified by devices such as the MegiQ RMS-0460 and Diamond Engineering D6050, are primarily designed for research and industrial applications. While they offer high accuracy and advanced features, their significant acquisition cost, operational complexity, and infrastructural requirements generally hinder their adoption within educational institutions that prioritise simplicity and affordability.

In order to overcome these barriers, various custom and open-source solutions have been presented in the literature. For example, Picco and Martin [2] proposed a low-cost setup based on modified Wi-Fi hardware and open-source firmware capable of measuring RF signals at 2.4 GHz. Their proposal includes an automated antenna positioning using stepper motors integrated with LabVIEW software.

Junfithrana et al.[3] and Simanjuntak et al.[4] describe the development of automated two-axis measurement systems controlled by microcontrollers and integrated with network analysers and Labview. Their aim was to demonstrate how accessible electronic components can be integrated into a measurement apparatus for educational purposes.

Additionally, open-source and software-defined radio (SDR) based platforms have broadened the scope for flexibility and adaptability in academic environments. Projects such as those in the Weber State University leverage SDRs, Arduino development boards, and GNU Radio to promote accessible measurement capabilities [5, 6]. Hardware improvements, including substituting PVC with 3d-printed parts and adopting modular assemblies, have contributed to enhanced portability and ease of customisation. Concurrent software advancements focusing on data processing, noise reduction, and error mitigation [7] support more reliable experimental outcomes and collaborative student research across diverse fields.

All of these works served as inspiration, but none of them met the design criteria that we would like a system of this nature, completely open-source, to have. Indeed, this work builds upon this foundation, offering a low-cost, computerised two-axis measurement system that complements theoretical instruction

with real-world experimental interaction. By facilitating interactive visualisation and analysis of antenna radiation patterns, the system provides an alternative pedagogical resource suitable for a variety of teaching and learning scenarios.

In the remaining sections of this paper, the antenna radiation measurement system will be explained, taking into consideration its distinct features with special focus on its mechatronic details.

3 Materials and Methods

As discussed previously, commercial antenna radiation pattern measurement systems, while highly accurate, are often prohibitively expensive and complex for educational environments. To address these challenges, a low-cost, replicable system was developed to enable didactic and experimental measurement of antenna radiation patterns. This section outlines the materials used, system configuration, and the methods adopted for calibration and validation.

3.1 System Description

The measurement system developed employs a structure with two rotational axes, enabling automatic scanning of the radiation pattern for antennas mounted on its upper support. The azimuth axis, located at the base, allows rotation from 0 to 360° in the horizontal plane. Above it, a perpendicular axis enables rotation in the orthogonal plane, also covering up to 360°. Depending on the antenna mounting orientation, this perpendicular axis may be adapted to scan the elevation axis. This configuration provides flexible and computer-controlled angular positioning of the antenna, allowing for antenna characterisation from different perspectives.

The system architecture is modular and structured into three primary subsystems: mechanical, electrical and software. The former includes all the rotational assemblies, supports, brackets, and transmission mechanisms. The transmission mechanisms, comprised of gears and drive belts, are propelled by NEMA 17 stepper motors. The electric signals used to move those motors are generated by a power driver, in this case a DRV8825, controlled through an Arduino Uno development board. The electrical subsystem also includes Hall-effect sensors for global system positioning and an RF power sensor based on the AD8313 integrated circuit. The third subsystem includes the firmware that lives in the microcontroller and the graphical interface developed for MATLAB[3]. Each one of those three subsystems will be further detailed in the next sections.

3.2 Mechanical Subsystem

From a mechanical point of view, the system can be seen as consisting of three parts. An upper part, responsible for holding the antenna, and able to tilt along

[3] A campus version of this software is licensed to the Polytechnic Institute of Bragança.

its axis. Figure 2 illustrates this part modelled in SolidWorks[4] CAD software. The two distinct rotational movements are highlighted: azimuth (blue arrow) and perpendicular rotation (red arrow). The transmission mechanism inside this part has been specially designed so as not to include any metal parts. In fact, all of the existing gears and bearings that support the main shaft are made of nylon. This aims to minimise any pattern distortion that can be caused by metallic parts close to the antenna.

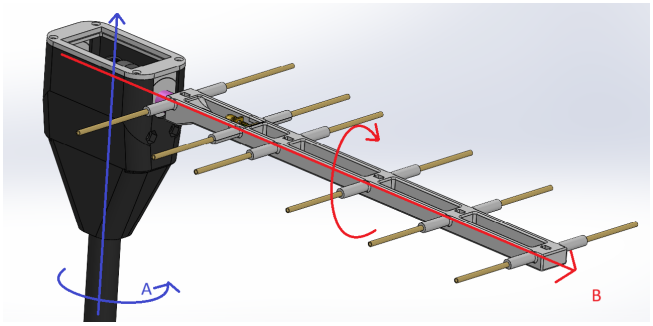


Fig. 2: SolidWorks rendering of the antenna measurement system highlighting the rotational movements: (A) azimuth rotation and (B) perpendicular axis rotation.

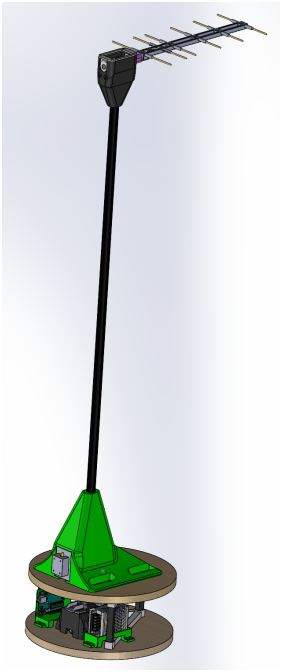
It is worth noticing that, even if Figure 2 depicts a particular type of antenna, the platform is intended for experiments with various antenna types. A quick-change bracket was designed to enable the fast replacement of antennas.

After this top antenna handler part, follows a central part consisting of a one-meter hollow fibreglass tube and a mechanical transmission system. The material used for the tube was carefully chosen to be non-conductive but, at the same time, robust and resilient to harsh environmental conditions. The tube length was selected such that it corresponds to three wavelengths for the band of interest.

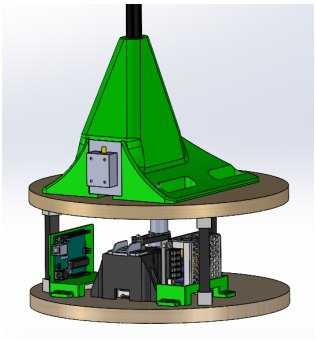
The tilt mechanism on the top part is driven by pulleys and a belt that runs inside the fibreglass tube. The mechanical actuation is provided by a stepper motor housed in the central part. At the same time, this central part, which is suspended by a system of ball bearings, is able to rotate, propelled by a gear mechanism located at the bottom part. This bottom part, where all the main electronic parts are housed, is made from wood-based chipboards cut in a circular shape with a 30 cm diameter. Figure 3 illustrates each one of the three mechanical parts.

Many of the fittings and stands were CAD-designed and 3d printed using polylactic acid (PLA).

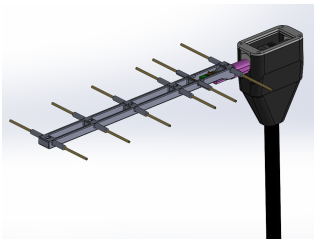
[4] Educational version licensed to the Polytechnic Institute of Bragança.



(a) Full system.



(b) Bottom section.



(c) Top section with test antenna.

Fig. 3: SolidWorks rendering of the antenna radiation pattern measurement system.

3.3 Electrical Subsystem

This section deals with the electrical subsystem, which is illustrated by the block diagram of Figure 4.

All the electrical power required to drive the system is provided by a 12 V, 2 A, AC to DC switch-mode power supply. The output voltage is connected to the local linear voltage regulator in the Arduino board and to the stepper motor driver. The stepper driver is based on the DRV8825 integrated circuit which is able to operate with voltages that span from 8 V to 45 V and is capable of driving up to 2.5 A for each of its two outputs. Among its many features, it includes several protection mechanisms, such as overcurrent and thermal shutdown, as well as the possibility of operating directly with microsteps. This driver is used for open-loop control of two NEMA 17 stepper motors. Those are two-phase motors presenting a 1.8 ° step angle and around 0.5 Nm of holding torque. The reference position of each stepper motor is provided by a Hall-effect sensor, which detects the magnetic field of a permanent magnet coupled to each motor shaft.

Finally, the RF power harvested by the antenna (DUT) is measured by the Analog Device's AD8313 chip. This integrated circuit behaves as a multistage demodulating logarithmic amplifier and is able to generate a DC voltage propor-

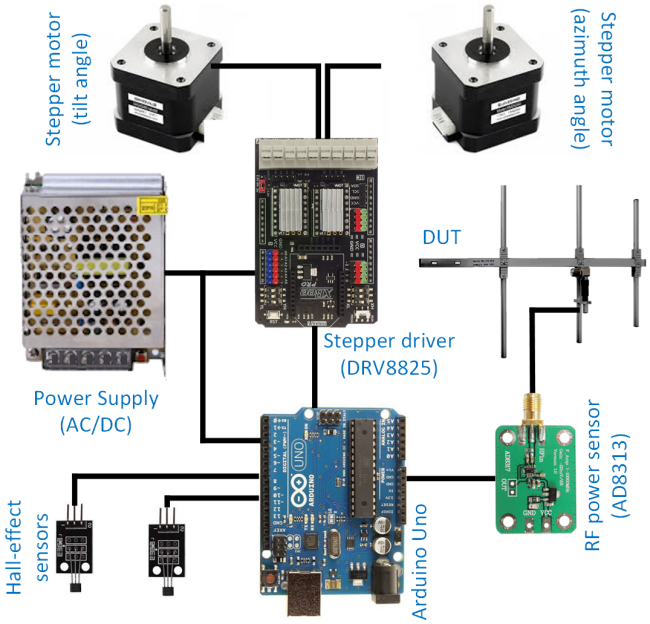


Fig. 4: Block diagram of the electronic modules and their connection.

tional to the input RF power. This device has a bandwidth that extends from 100 MHz to 2.5 GHz with a total dynamic range of 70 dB.

3.4 Software Subsystem

System control and automation are achieved through embedded firmware running on the Arduino UNO platform. The firmware, developed in C++, controls the movement of both stepper motors (azimuth and perpendicular axes), executes homing routines, and gathers and filters data delivered by the AD8313 sensor.

At a higher level, a MATLAB program takes care of the data acquisition process and user interface. Communication between MATLAB and the microcontroller is established through a serial connection over USB. MATLAB sends motion commands and data requests to the microcontroller. The software application also handles homing routines and manages retransmission in case of communication errors. Acquired data are stored and immediately presented as polar plots, enabling straightforward visualisation of radiation patterns for analysis and educational use.

3.5 System Operation

When initialised, the system begins by running a homing calibration. During this stage, Hall-effect sensors allow the system to automatically detect the reference

point (zero) for each axis. The stepper motor's driver was set to operate in micro-stepping mode (1/16). A transmitting antenna is operating as a beacon, transmitting at a given frequency.

Data provided by the RF power sensor is acquired at a user-defined angular interval and sent to MATLAB to be processed and visualised. The entire measurement and data acquisition flow, along with system interconnections, is summarised by Figure 5.

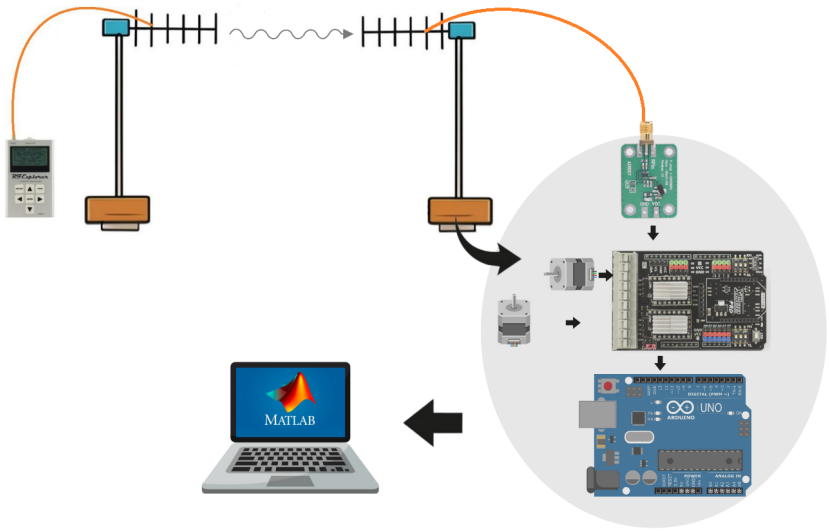


Fig. 5: Schematic showing the flow of the signal from generation to analysis.

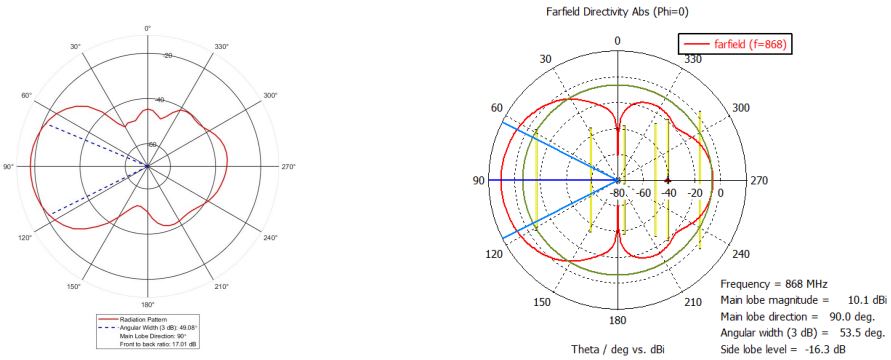
4 Results and Discussion

To validate the antenna radiation pattern measurement system, a Yagi-Uda antenna, to operate at 868 MHz, was sized and built. During this stage, simulation was performed using CST Studio Suite[5]. Those simulations would serve as a term of comparison for the results obtained experimentally.

Experimental measurements were performed along the azimuthal plane with a 5 ° resolution. Figure 6 presents a comparison between the measured radiation pattern and the simulated pattern obtained. The measured data, shown in Figure 6a, were obtained by averaging 100 samples per angular interval. The simulated pattern, depicted in Figure 6b, serves as a theoretical reference for validation.

As shown in Figure 6, both the measured and simulated patterns exhibit a main lobe direction at 90°, demonstrating that the system is able to determine

[5] The free educational version of this software was used in the simulations.



(a) Real data acquired (b) Simulation of the radiation pattern

Fig. 6: Comparison of real and simulated radiation pattern.

the direction of the main lobe. The measured beamwidth (-3 dB) of 49.08° is closely aligned with the simulated value of 53.5°, front-to-back ratio is 17.01 dB while the simulated value is 16.31 dB, a comparison not shown in the figure. These results suggest that the system can capture the main features of the radiation pattern, including the main lobe and secondary lobes.

Some discrepancies were observed between the measured and simulated patterns, which can be attributed to several factors. The measurement environment, although selected to minimise reflective effects, could not eliminate all sources of interference, unlike the free-space conditions assumed in the simulation. Additionally, the antenna may have captured signals at frequencies other than the intended 868 MHz, and the presence of other transmitting devices in the vicinity likely introduced noise and interference. Such effects could be mitigated in a controlled setting, such as an anechoic chamber, which was not available for this study. These environmental factors highlight the challenges of real-world measurements compared to the idealised conditions of the simulation, yet the system's ability to produce consistent and reproducible results under these conditions demonstrates its robustness for educational purposes.

Overall, the results demonstrate that the system is capable of capturing key radiation pattern features, with measurements that generally agree with the simulation. While environmental limitations affected absolute accuracy, the system remains a robust and versatile platform for practical education in telecommunications. These results support the suitability of the system as an alternative learning resource for exploring fundamental concepts in antenna behaviour, especially in academic and training environments where cost is a limiting factor.

5 Conclusions and Future Work

In the course of this article, the design, implementation and testing of a system that allows the measurement of the radiation pattern of antennas was docu-

mented. Although these types of devices exist commercially, their price makes them quite inaccessible for use in an educational context.

The designed system has two degrees of freedom, allowing the automatic acquisition of the radiation pattern of an antenna in 360 degrees.

Preliminary tests carried out along the azimuthal plane suggest that the presented methodology provides a close match to those obtained by simulation.

Future improvements include the addition of bandpass filters to suppress out-of-band signals, the addition of a balun to mitigate transmission line imbalance, and the improvement of impedance matching.

Additional trials using different antenna types and along different planes will be performed. Once thoroughly tested, the entire project will be placed in a repository to promote its replication. Moreover, interactive learning modules, aiming to increase the educational engagement with the subject of RF engineering and antenna theory, will be developed.

Acknowledgment

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